Chapter 23: Electromagnetic Induction, AC Circuits, and Electrical Technologies

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Lecture 23

Part I

Electromagnetic Induction and Magnetic Flux

Outline of Part I: Electromagnetic Induction and Magnetic Flux

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Faraday’s Discovery

Magnetic Flux

Faraday’s Law and Lenz’s Law

What is Electromagnetic Induction?

Electromagnetic induction is the production of an emf — and possibly a current — by a *changing* magnetic environment.

Faraday's Key Discovery (1831)

- A **stationary** magnetic field produces *no* induced emf
- A **changing** magnetic field induces an emf
- Relative motion between conductor and field source is sufficient
- The induced emf can drive current in a closed circuit

Central Principle:

Only *changing* magnetic flux induces emf — not the flux itself.

$$\Delta\Phi \neq 0 \;\Rightarrow\; \varepsilon \neq 0$$

Outline of Part I: Electromagnetic Induction and Magnetic Flux

Faraday's Discovery

Magnetic Flux

Faraday's Law and Lenz's Law

Magnetic Flux Φ

Magnetic Flux — SI Unit: weber $(\mathrm{Wb} = \mathrm{T} \cdot \mathrm{m}^2)$

Magnetic flux measures how much of the magnetic field *passes through* a surface:

$$\boxed{\Phi = BA\cos\theta}$$

where θ is the angle between $\vec{B}$ and the **surface normal** (not the surface plane itself).

Variable Definitions

- Φ: magnetic flux (Wb)
- B: magnetic field magnitude (T)
- A: loop area (m^2)
- θ: angle between $\vec{B}$ and area normal

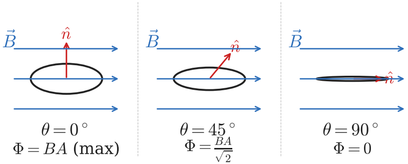
Special Cases

- $\theta = 0°$: $\Phi = BA$ (maximum)
- $\theta = 90°$: $\Phi = 0$ (zero flux)

Magnetic Flux: Geometric Interpretation

Three Orientations of the Same Loop

- $\vec{B}$ perpendicular to loop ($\theta = 0°$): **maximum** flux
- $\vec{B}$ at 45° to normal ($\theta = 45°$): **reduced** flux
- $\vec{B}$ parallel to loop surface ($\theta = 90°$): **zero** flux



Common Mistake

θ is between $\vec{B}$ and the **normal**, not the plane.

When $\vec{B}$ is parallel to the loop plane, $\theta = 90°$ and $\Phi = 0$.

Check Your Understanding

A conducting loop lies flat on a table. A uniform magnetic field $\vec{B}$ points straight up (perpendicular to the table).

(a) What is the flux through the loop?

(b) The loop is tilted $90°$ so it stands vertical. What is the flux now?

Come to lecture to see the answer.

Outline of Part I: Electromagnetic Induction and Magnetic Flux

Faraday's Law of Induction

Faraday's Law — SI Unit: volt (V)

The magnitude of the induced emf equals the rate of change of magnetic flux through a coil of N turns:

$$\boxed{\varepsilon = -N\frac{\Delta\Phi}{\Delta t}}$$

Physical Interpretation

- Larger $\Delta\Phi \Rightarrow$ larger emf
- Faster change $\Rightarrow$ larger emf
- More turns $N \Rightarrow$ each adds emf in series
- Steady flux: $\Delta\Phi = 0 \Rightarrow \varepsilon = 0$

The Negative Sign

The "$-$" sign encodes **Lenz's Law**:

The induced emf always *opposes* the change that produced it.

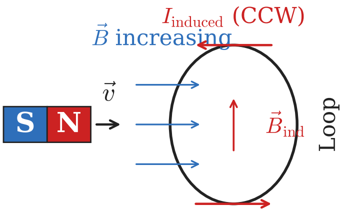
Lenz's Law: Direction of Induced Current

Lenz's Law

The induced current creates a magnetic field that **opposes the change in flux** that caused it.

Examples

- Flux **increasing** into page ⇒ induced current creates field *out of* page
- Flux **decreasing** ⇒ induced current tries to maintain original flux
- Magnet approaching ⇒ loop repels the magnet
- Magnet receding ⇒ loop attracts the magnet



Three-Step Induction Reasoning Framework

Step 1. **Identify** the magnetic flux through the loop.

Step 2. **Determine** whether the flux is increasing or decreasing.

Step 3. **Find** the current direction that opposes the change (right-hand rule).

Applied to: North Pole Approaching Loop

1. Flux through loop is directed toward the north pole
2. Magnet approaches $\Rightarrow$ flux **increases**
3. Induced current must create field pointing *away* from the magnet $\Rightarrow$ counterclockwise current (viewed from magnet)

Use this same framework for:

- moving magnets
- changing currents
- rotating coils
- moving conductors (motional emf)

Example: Induced EMF from Changing Field

Induced EMF in a Circular Loop

A single-turn circular loop of radius $r = 6.00\,\text{cm}$ is perpendicular to a magnetic field that increases from $5.00\,\text{T}$ to $25.0\,\text{T}$ in $\Delta t = 0.100\,\text{s}$. Find $|\varepsilon|$.

$$A = \pi r^2 = \pi(0.0600)^2 = 1.131 \times 10^{-2}\,\text{m}^2$$

$$\Delta\Phi = A\,\Delta B = (1.131 \times 10^{-2})(20.0) = 0.226\,\text{Wb}$$

$$|\varepsilon| = N\frac{|\Delta\Phi|}{\Delta t} = \frac{0.226}{0.100}$$

$$\boxed{|\varepsilon| = 2.26\,\text{V}}$$

The emf scales with area: a larger loop would intercept more flux change.

With N turns, emf would be $N \times 2.26\,\text{V}$.

- What if Δt were halved? The emf would double.
- What if the loop were tilted $30°$? $\Delta\Phi$ would be scaled by $\cos 30°$.

Check Your Understanding

A north magnetic pole approaches a conducting loop from the left.

Using the three-step framework: what direction does the induced current flow (viewed from the left)?

Come to lecture to see the answer.

Part II

Motional EMF and Eddy Currents

Outline of Part II: Motional EMF and Eddy Currents

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Motional EMF

Eddy Currents

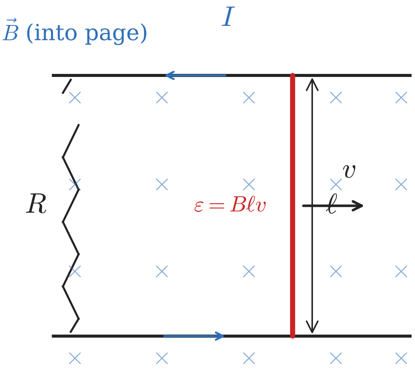
Motional EMF: Physical Origin

Charge Separation by Magnetic Force

A conductor moving through a magnetic field experiences:

$$\vec{F} = q\vec{v} \times \vec{B}$$

- Positive charges pushed to one end
- Negative charges accumulate at the other end
- This charge separation creates a voltage (emf)
- Moving conductor sweeps area $\Rightarrow \Delta\Phi \neq 0$



Motional EMF Equation

Motional EMF — SI Unit: volt (V)

When $\vec{B}$, conductor length ℓ, and velocity $\vec{v}$ are mutually perpendicular:

$$\boxed{\varepsilon = B\ell v}$$

Consistency with Faraday's Law

As rod moves distance $\Delta x = v\,\Delta t$:

$$\Delta\Phi = B\,\ell\,\Delta x = B\ell v\,\Delta t$$

$$\varepsilon = \frac{\Delta\Phi}{\Delta t} = B\ell v \checkmark$$

Condition

- Requires $\vec{v} \perp \vec{B}$ and $\vec{v} \perp \vec{\ell}$
- If $\vec{v} \parallel \vec{B}$: no force on charges, $\varepsilon = 0$

Example: Motional EMF of a Sliding Rod

Sliding Conducting Rod

A 0.300 m conducting rod slides along rails in field $B = 0.800\,\mathrm{T}$ perpendicular to the rod. The rod moves at $v = 2.00\,\frac{\mathrm{m}}{\mathrm{s}}$. Find (a) the induced emf, and (b) the current if the circuit resistance is $R = 0.500\,\Omega$.

(a) $$\varepsilon = B\ell v$$
$$= (0.800)(0.300)(2.00)$$
$$\boxed{\varepsilon = 0.480\,\mathrm{V}}$$

(b) $$I = \frac{\varepsilon}{R}$$
$$= \frac{0.480}{0.500}$$
$$\boxed{I = 0.960\,\mathrm{A}}$$

Power delivered: $P = \varepsilon I = 0.461\,\mathrm{W}$.

This power comes from the **mechanical work** done pushing the rod — the induced current creates a drag force opposing the motion (Lenz's Law).

- If v triples: emf triples, current triples, power increases $9\times$.
- Who supplies the energy? Whoever pushes the rod.

Check Your Understanding

A conducting rod moves through a magnetic field **parallel** to $\vec{B}$ (i.e. $\vec{v} \,||\, \vec{B}$).

What is the induced emf? Explain why.

Come to lecture to see the answer.

Outline of Part II: Motional EMF and Eddy Currents

Motional EMF

Eddy Currents

Eddy Currents in Bulk Conductors

What Are Eddy Currents?

When a bulk conductor moves through a magnetic field (or a changing field passes through it), Faraday's Law produces **circulating current loops** inside the material.

Magnetic Damping

- Eddy currents obey Lenz's Law: they oppose the change causing them
- This creates a **magnetic drag force** opposing motion
- Larger velocity $\Rightarrow$ stronger eddy currents $\Rightarrow$ larger drag
- Energy dissipated as **Joule heating**

Reducing eddy currents:

Laminated cores (thin insulated sheets) break up current paths.

Slotting a plate disrupts loops — less damping, more energy retained.

Applications of Eddy Currents

Desired Eddy Current Effects

- **Induction cooktops**: AC field creates eddy currents in pot, heating food directly
- **Magnetic braking**: roller coasters, MRI table brakes, train systems
- **Metal detectors**: eddy currents in hidden metal alter the detector's field
- **Electromagnetic damping**: laboratory balances

Unwanted Eddy Currents

- Power losses in transformer cores
- Heating in motor cores
- Solution: **laminated silicon steel** cores

Check Your Understanding

A solid aluminum plate and an identical plate with many thin slots are both dropped between the poles of a strong magnet.

Which plate falls more slowly, and why?

Come to lecture to see the answer.

Part III

Generators, Motors, and Transformers

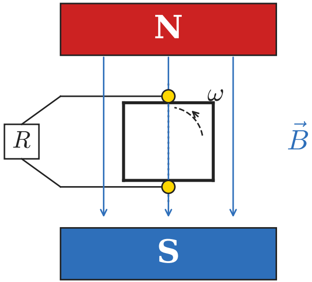
Outline of Part III: Generators, Motors, and Transformers

Outline of Part III: Generators, Motors, and Transformers

How an AC Generator Works

Rotating Coil in a Magnetic Field

- A coil of wire rotates in a uniform magnetic field
- As it rotates, the flux through the coil changes sinusoidally
- By Faraday's Law, a sinusoidal emf is induced
- Slip rings connect the rotating coil to the external circuit



$\varepsilon_0 = NAB\omega$

Energy Conversion

Mechanical work (turbine, engine) $\rightarrow$ changing flux $\rightarrow$ induced emf $\rightarrow$ electrical energy.

This is the three-step induction framework applied continuously.

Generator EMF Equation

AC Generator Output — SI Unit: volt (V)

For a coil of N turns, area A, rotating at angular frequency ω in field B:

$$\boxed{\varepsilon(t) = NAB\omega \sin(\omega t) = \varepsilon_0 \sin(\omega t)}$$

where $\varepsilon_0 = NAB\omega$ is the **peak emf**.

Increasing Peak EMF

- More turns N
- Larger coil area A
- Stronger field B
- Faster rotation ω

Flux vs. EMF

EMF is maximum when *flux* is zero.

EMF is zero when *flux* is maximum.

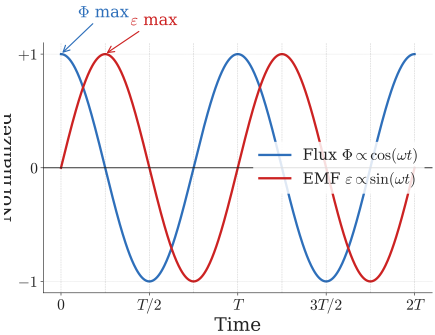
EMF $\propto$ *rate* of flux change, not the flux itself!

Flux and EMF: Phase Relationship

Generator Graphs

- Flux: $\Phi(t) = NBA\cos(\omega t)$ (cosine)
- EMF: $\varepsilon(t) = NAB\omega\sin(\omega t)$ (sine)
- EMF is the **negative time-derivative** of flux
- They are 90° out of phase

When coil is $\perp$ to $\vec{B}$: flux is maximum *but* not changing — emf $= 0$.
When coil is $\|$ to $\vec{B}$: flux is zero *but* changing fastest — emf is maximum.



Example: Average EMF from a Rotating Coil

Rotating Circular Coil

A 200-turn circular coil of radius $r = 5.00\,\text{cm}$ is in field $B = 1.25\,\text{T}$ and rotates from $\theta = 0°$ to $90°$ in $\Delta t = 15.0\,\text{ms}$. Find the average induced emf.

$$A = \pi r^2 = \pi(0.05)^2 = 7.854 \times 10^{-3}\,\text{m}^2$$

$$\Phi_i = BA\cos 0° = (1.25)(7.854 \times 10^{-3})$$

$$= 9.817 \times 10^{-3}\,\text{Wb}$$

$$\Phi_f = BA\cos 90° = 0$$

$$|\varepsilon| = \frac{N|\Delta\Phi|}{\Delta t}$$

$$= \frac{200 \times 9.817 \times 10^{-3}}{0.0150}$$

$$\boxed{|\varepsilon| \approx 131\,\text{V}}$$

A 200-turn coil rotating a quarter turn in 15 ms produces 131 V.

Real generators increase N, A, B, and spin speed to produce kilovolts.

Check Your Understanding

A generator coil is oriented so that the magnetic flux through it is at its **maximum value** at $t = 0$.

At this instant, is the induced emf at its maximum, or is it zero? Explain.

Come to lecture to see the answer.

Outline of Part III: Generators, Motors, and Transformers

Back EMF: Motors Are Also Generators

What is Back EMF?

As a motor's coil spins, it acts as a generator and produces an emf that **opposes** the applied voltage. This is the **back emf** ε_b.

Net Current in a Motor

$$I = \frac{V - \varepsilon_b}{R}$$

where V is applied voltage, R is coil resistance.

Motor Stall!

If rotation stops: $\varepsilon_b \to 0$, current $I \to V/R$ — enormous, causing overheating.

Startup: No rotation $\Rightarrow$ no back emf $\Rightarrow$ very large current.

Operating speed: Back emf grows $\Rightarrow$ net current falls.

Starter resistors or reduced voltage limit startup current.

Example: Back EMF Calculation

Motor Back EMF

A DC motor connected to $V = 120\,\text{V}$ has coil resistance $R = 0.800\,\Omega$. At full operating speed it draws $I = 8.00\,\text{A}$. Find the back emf ε_b.

$$I = \frac{V - \varepsilon_b}{R}$$

$$\varepsilon_b = V - IR$$

$$= 120 - (8.00)(0.800)$$

$$\boxed{\varepsilon_b = 113.6\,\text{V}}$$

Back emf of $113.6\,\text{V}$ nearly equals the $120\,\text{V}$ supply.

At startup ($\varepsilon_b = 0$):

$I_{start} = 120\,\text{V}/0.800\,\Omega = 150\,\text{A}$

— nearly $19\times$ the running current!

- How does loading the motor affect ε_b?
- Why do motors need overload protection?

Check Your Understanding

A motor is running steadily when its load suddenly doubles, slowing the motor down.

What happens to (a) the back emf, (b) the net current, and (c) the power consumed?

Come to lecture to see the answer.

Outline of Part III: Generators, Motors, and Transformers

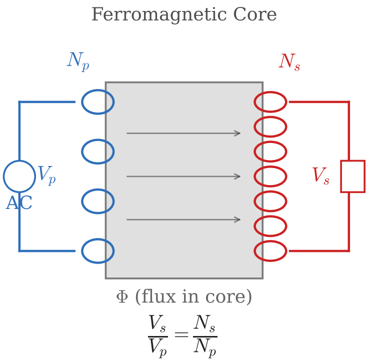
How Transformers Work

Mutual Induction in a Ferromagnetic Core

- AC current in the **primary coil** creates a changing magnetic field
- The ferromagnetic core **confines and channels** the flux
- Changing flux passes through the **secondary coil**
- By Faraday's Law, an emf is induced in the secondary

Requires AC!

DC produces constant flux: $\Delta\Phi/\Delta t = 0$, so no sustained emf in secondary.



Ideal Transformer Equations

Transformer Relations (Ideal)

$$\boxed{\frac{V_s}{V_p} = \frac{N_s}{N_p}} \quad \text{and} \quad \boxed{V_p I_p = V_s I_s}$$

where subscripts p and s denote primary and secondary.

Voltage and Current Rules

- **Step-up**: $N_s > N_p$, $V_s > V_p$, $I_s < I_p$
- **Step-down**: $N_s < N_p$, $V_s < V_p$, $I_s > I_p$
- Power is **conserved**: $P_p = P_s$ (ideal)
- Higher voltage $\Rightarrow$ lower current

Power Grid Application

Power lines transmit at **high voltage** (low current) to minimize I^2R losses.

Step-down transformers near homes reduce to safe 120 V/240 V.

Example: Transformer Voltage Ratio

Step-Up Transformer

A transformer has $N_p = 40$ turns and $N_s = 600$ turns. (a) If $V_p = 120\,\text{V}$, find V_s. (b) If $I_p = 5.00\,\text{A}$, find I_s.

$$\text{(a)} \quad V_s = V_p \frac{N_s}{N_p} = 120 \cdot \frac{600}{40}$$

$$\boxed{V_s = 1800\,\text{V}}$$

$$\text{(b)} \quad I_s = I_p \frac{N_p}{N_s} = 5.00 \cdot \frac{40}{600}$$

$$\boxed{I_s = 0.333\,\text{A}}$$

Check power conservation:

$$P_p = (120)(5.00) = 600\,\text{W}$$

$$P_s = (1800)(0.333) \approx 600\,\text{W} \checkmark$$

Voltage increased by factor 15; current decreased by factor 15.

Check Your Understanding

Electrical power is transmitted over long distances at very high voltages (e.g. 500 kV).

Why does high voltage reduce power losses in the transmission lines?

Come to lecture to see the answer.

Part IV

Inductance

Outline of Part IV: Inductance

Outline of Part IV: Inductance

Self-Inductance

Self-Inductance L — SI Unit: henry $(\mathrm{H} = \mathrm{V} \cdot \mathrm{s}/\mathrm{A})$

When the current through a coil changes, it induces an emf in itself:

$$\boxed{\varepsilon = -L\frac{\Delta I}{\Delta t}}$$

The self-inductance L quantifies how strongly a coil resists current changes.

Physical Interpretation

- Changing current $\Rightarrow$ changing magnetic field
- Changing field $\Rightarrow$ changing flux in coil
- By Faraday's Law: emf induced opposing the current change
- Inductors resist **changes** in current, not current itself

So-called Electromagnetic Inertia

Just as mass resists changes in velocity, inductance resists changes in current.

L depends only on geometry, not on current.

Solenoid Inductance and Stored Energy

Inductance of a Solenoid

$$\boxed{L = \frac{\mu_0 N^2 A}{\ell}}$$

where N is turns, A is cross-sectional area, ℓ is length.

Energy Stored in an Inductor

$$\boxed{E_{\text{ind}} = \frac{1}{2} L I^2}$$

The energy is stored in the **magnetic field** inside the inductor — analogous to $PE_{el} = \frac{1}{2}kx^2$ stored in a spring's elastic field.

Increasing N, A, or using a ferromagnetic core increases L.

Longer solenoid $\Rightarrow$ smaller L for same N.

Example: Energy Stored in a Solenoid

Solenoid Inductance and Energy

A solenoid has $N = 100$ turns, length $\ell = 0.150\,\text{m}$, area $A = 2.00 \times 10^{-4}\,\text{m}^2$. (a) Find L. (b) Find the stored energy when carrying $I = 2.00\,\text{A}$.

$$\text{(a)} \quad L = \frac{\mu_0 N^2 A}{\ell}$$

$$= \frac{(4\pi \times 10^{-7})(100)^2(2.00 \times 10^{-4})}{0.150}$$

$$\boxed{L \approx 16.8\,\mu\text{H}}$$

$$\text{(b)} \quad E = \tfrac{1}{2} L I^2$$

$$= \tfrac{1}{2}(1.68 \times 10^{-5})(2.00)^2$$

$$\boxed{E \approx 3.35 \times 10^{-5}\,\text{J}}$$

Small solenoids store tiny amounts of energy.

Superconducting MRI magnets ($L \sim 10\,\text{H}$, $I \sim 200\,\text{A}$) store $\sim 200\,\text{kJ}$.

Check Your Understanding

The current through an inductor is doubled.

By what factor does the stored energy change?

Come to lecture to see the answer.

Outline of Part IV: Inductance

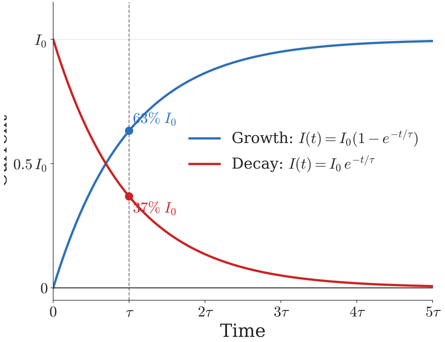
Self-Inductance

RL Circuits

RL Circuit Transient Behavior

What Happens When the Switch Closes?

- At $t = 0$: inductor opposes sudden current change $\Rightarrow I = 0$
- As t increases: current **rises gradually**, following an exponential
- At $t \to \infty$: $I \to I_0 = V/R$ (inductor behaves like a wire)



Intuition: Electromagnetic Inertia

Building current requires building a magnetic field. This takes time and work. The inductor opposes the change just as mass resists acceleration.

RL Time Constant and Current Equations

RL Time Constant

$$\boxed{\tau = \frac{L}{R}}$$

Current Growth and Decay

$$\text{Growth:} \quad I(t) = I_0\left(1 - e^{-t/\tau}\right)$$

$$\text{Decay:} \quad I(t) = I_0\, e^{-t/\tau}$$

Interpreting τ

- At $t = \tau$: current reaches 63% of I_0 (growth)
- At $t = \tau$: current decays to 37% of I_0 (decay)
- After 5τ: essentially complete
- Larger L or smaller $R \Rightarrow$ slower response

At $t = 0$: current cannot change instantaneously.

At $t \to \infty$: inductor looks like a short circuit.

Opposite behavior from capacitors!

Example: RL Time Constant

RL Circuit with Switch

An RL circuit has $L = 7.50\,\text{mH}$ and $R = 3.00\,\Omega$, connected to $V = 12.0\,\text{V}$. (a) Find τ. (b) Find the current at $t = 1.00\,\text{ms}$.

$$\text{(a)} \quad \tau = \frac{L}{R} = \frac{7.50 \times 10^{-3}}{3.00}$$

$$\boxed{\tau = 2.50\,\text{ms}}$$

$$\text{(b)} \quad I_0 = \frac{V}{R} = \frac{12.0}{3.00}$$

$$= 4.00\,\text{A}$$

$$I(t) = I_0\left(1 - e^{-t/\tau}\right)$$

$$= 4.00\left(1 - e^{-1.00/2.50}\right)$$

$$\boxed{I \approx 1.37\,\text{A}}$$

At $t = \tau = 2.50\,\text{ms}$:

$I = 4.00(1 - e^{-1}) = 2.53\,\text{A}$

(63% of I_0).

At $t = 1.00\,\text{ms}$ (before τ), we are at only 34% of I_0.

Check Your Understanding

An RL circuit has $L = 10.0\,\mathrm{mH}$ and $R = 5.00\,\Omega$.

(a) What is the time constant τ?

(b) At the instant the switch closes, what is the current through the inductor?

Come to lecture to see the answer.

Part V

AC Circuits

Outline of Part V: AC Circuits

AC Reactance

RLC Circuits and Resonance

Outline of Part V: AC Circuits

AC Reactance

RLC Circuits and Resonance

Inductive Reactance

Inductive Reactance X_L — SI Unit: ohm (Ω)

An inductor in an AC circuit opposes current with:

$$\boxed{X_L = 2\pi f L}$$

Frequency Dependence

- **Higher** f: current changes faster $\Rightarrow$ larger opposing emf $\Rightarrow$ X_L increases
- **Lower** f **(DC)**: $f = 0 \Rightarrow X_L = 0$ (inductor looks like a plain wire)
- Inductor: **low-pass** behavior (passes low f, blocks high f)

Phase Relationship

In an inductor, voltage **leads** current by $90°$:

Current lags because inductance resists rapid current changes.

Capacitive Reactance

Capacitive Reactance X_C — SI Unit: ohm (Ω)

A capacitor in an AC circuit opposes current with:

$$\boxed{X_C = \frac{1}{2\pi f C}}$$

Frequency Dependence

- **Higher** f: capacitor charges/discharges faster $\Rightarrow$ X_C decreases
- **Lower** f **(DC)**: $f \to 0 \Rightarrow X_C \to \infty$ (capacitor blocks DC)
- Capacitor: **high-pass** behavior (passes high f, blocks low f)

Phase Relationship

In a capacitor, current **leads** voltage by 90°:

Charge builds up *after* current flows.

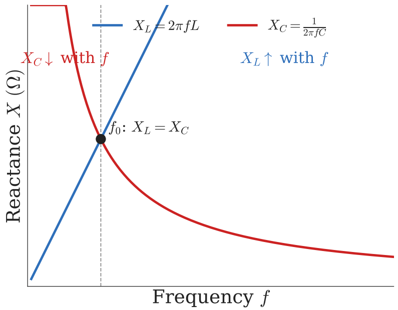
Reactance vs. Frequency: Comparison

Opposite Frequency Behaviors

- $X_L = 2\pi f L$: **increases** with frequency
- $X_C = 1/(2\pi f C)$: **decreases** with frequency
- They cross at a specific frequency (related to resonance)

Filtering Applications

- **Inductor**: blocks high-f, passes low-f (low-pass filter)
- **Capacitor**: blocks low-f, passes high-f (high-pass filter)
- **Crossover networks**: route bass to woofer, treble to tweeter



Check Your Understanding

The frequency of an AC source is doubled.

(a) What happens to X_L?

(b) What happens to X_C?

(c) Which component is easier to pass current through at high frequency?

Come to lecture to see the answer.

Outline of Part V: AC Circuits

Impedance in RLC Series Circuits

Impedance Z — SI Unit: ohm (Ω)

In a series RLC circuit, R, X_L, and X_C combine using the Pythagorean theorem (they are 90° out of phase):

$$\boxed{Z = \sqrt{R^2 + (X_L - X_C)^2}}$$

AC Ohm's Law: $I_{\text{rms}} = \dfrac{V_{\text{rms}}}{Z}$

Key Features

- $X_L > X_C$: circuit is **inductive** (voltage leads current)
- $X_C > X_L$: circuit is **capacitive** (current leads voltage)
- $X_L = X_C$: $Z = R$ (purely resistive) — **resonance**

$Z \geq R$ always.

At resonance, impedance is minimized and current is maximized.

Resonance in RLC Circuits

Resonant Frequency f_0

Set $X_L = X_C$ and solve:

$$2\pi f_0 L = \frac{1}{2\pi f_0 C}$$

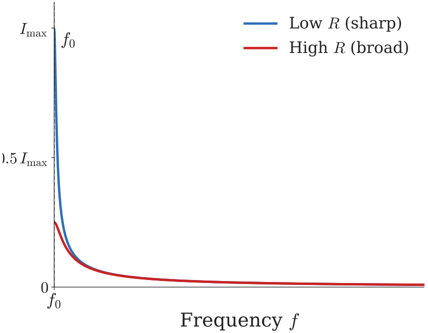
$$\boxed{f_0 = \frac{1}{2\pi\sqrt{LC}}}$$

Energy Picture at Resonance

Energy **oscillates** between:

- Electric field energy in capacitor: $\frac{1}{2}CV^2$
- Magnetic field energy in inductor: $\frac{1}{2}LI^2$

Like a pendulum swapping $PE \leftrightarrow KE$.



Example: RLC Resonant Frequency

Finding Resonant Frequency

A series RLC circuit has $R = 40.0\,\Omega$, $L = 1.00\,\text{mH}$, and $C = 0.500\,\mu\text{F}$. (a) Find f_0. (b) Find the impedance at resonance.

$$\text{(a)}\quad f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(1.00\times10^{-3})(0.500\times10^{-6})}}$$

$$\boxed{f_0 \approx 7117\,\text{Hz}}$$

$$\text{(b)}\quad X_L = X_C \text{ at resonance}$$

$$\boxed{Z = R = 40.0\,\Omega}$$

At resonance, the circuit acts as a pure resistor.

Reactive elements cancel exactly.

Adjusting C shifts f_0 — this is how radio tuners select stations.

Check Your Understanding

In a series RLC circuit at resonance, what is the impedance Z equal to?

Is the current at resonance at its maximum or minimum? Why?

Come to lecture to see the answer.

Part VI

Electrical Safety

Outline of Part VI: Electrical Safety

Outline of Part VI: Electrical Safety

Physiological Effects of Electric Current

Current Through the Body

- $\sim 1\,\mathrm{mA}$: just perceptible, tingling
- $\sim 5\,\mathrm{mA}$: maximum harmless for most people
- $\sim 10\,\mathrm{mA}$: muscle contraction, may not release grip
- $\sim 15\,\mathrm{mA}$: loss of muscular control
- $\sim 70\,\mathrm{mA}$: ventricular fibrillation possible
- $\sim 100\,\mathrm{mA}$: lethal if sustained $> 5\,\mathrm{s}$

Shock Danger

AC at 60 Hz is especially dangerous — it can disrupt the heart's rhythm.

The current path through the body matters: chest-to-chest is most dangerous.

It is current, not voltage, that injures.

But $I = V/R$ — high voltage drives high current through body resistance ($\sim 1000\,\Omega$ to $100\,\mathrm{k}\Omega$).

Fuses and Circuit Breakers

Overload Protection

- **Fuse**: thin wire melts when current exceeds rating — one-time device
- **Circuit Breaker**: bimetallic strip (heat) or electromagnetic trip opens circuit — resettable
- Both protect **wiring from overheating** and fire
- They do *not* protect against shock through leakage paths

Grounding

The third wire (ground) provides a low-resistance return path for fault currents — tripping the breaker before current travels through a person.

Household circuits: 15 A or 20 A breakers.

High-power circuits (dryer, oven): 30 A–50 A.

Ground Fault Circuit Interrupters (GFCIs)

How a GFCI Works

- Continuously compares current in hot wire vs. neutral wire
- If difference $> 5\,\text{mA}$: current is leaking through unintended path
- GFCI trips in $\sim 1/40\,\text{s}$ — fast enough to prevent electrocution
- Uses a small transformer to sense the current imbalance

Where GFCIs Are Required

Bathrooms, kitchens, outdoor outlets, garages, basements, near water.

GFCI vs. Circuit Breaker:

Breaker trips at 15–20 A — too high to protect a person.

GFCI trips at ∼5 mA — specifically designed for shock protection.

They protect different hazards; use both!

Check Your Understanding

A circuit breaker is rated at 20 A. A person accidentally touches a live wire and 100 mA flows through their body.

Will the circuit breaker protect them? Why or why not? What device would?

Come to lecture to see the answer.

Part VII

Chapter Synthesis

Outline of Part VII: Chapter Synthesis

The Unifying Principle

Outline of Part VII: Chapter Synthesis

The Unifying Theme of Chapter 23

Changing magnetic flux $\Rightarrow$ **induced emf** $\Rightarrow$ **induced current** $\Rightarrow$ **opposition to the change** (Lenz's Law)

Every Topic Follows This Chain

- **Generators**: rotation $\rightarrow$ changing flux $\rightarrow$ AC output
- **Motors**: current $\rightarrow$ torque, plus back emf opposes supply
- **Transformers**: AC current $\rightarrow$ changing flux $\rightarrow$ secondary emf
- **Inductors**: changing current $\rightarrow$ opposing self-induced emf
- **Eddy currents**: motion $\rightarrow$ changing flux $\rightarrow$ drag force

Technologies Powered by Induction

- Electric power grids
- MRI scanners
- Wireless charging (Qi)
- Induction cooktops
- Electric guitar pickups
- Metal detectors
- Radio tuning circuits

Key Equations of Chapter 23

Induction and Generation

$$\Phi = BA\cos\theta$$

$$|\varepsilon| = N\frac{|\Delta\Phi|}{\Delta t}$$

$$\varepsilon = B\ell v$$

$$\varepsilon_0 = NAB\omega$$

Transformers

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}, \qquad V_p I_p = V_s I_s$$

Inductance and RL

$$\varepsilon = -L\frac{\Delta I}{\Delta t} \qquad L = \frac{\mu_0 N^2 A}{\ell}$$

$$E_{\text{ind}} = \tfrac{1}{2}LI^2 \qquad \tau = \frac{L}{R}$$

AC Circuits

$$X_L = 2\pi f L, \quad X_C = \frac{1}{2\pi f C}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$